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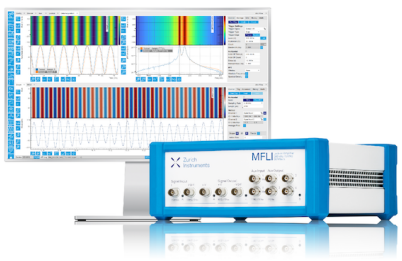
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Inverted perturbation approach (IPA) potentials and adiabatic corrections of the $X\,^1\Sigma^+$ state of the lithium hydrides near the dissociation limits

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Laser-induced fluorescence (LIF) spectra of the A–X system of the LiH isotopes, excited by the UV lines of an argon and a krypton ion laser, were observed and analyzed. Three long LIF series of different LiH isotopes were taken with a 3.4 m high resolution spectrograph. The results were combined with previous experimental data to study the breakdown of the Born–Oppenheimer approximation in the $X\,^1\Sigma^+$ electronic ground state of the isotopic LiH molecules. Using a quantum mechanical variational method, potential energy curves of the four LiH isotopes were extended almost up to the dissociation limits.

I. INTRODUCTION

Lithium hydride has been the object of intense theoretical and spectroscopic study since the 1930's. Its simple electronic structure makes the molecule one of the favorite subjects for testing different quantum-chemical techniques.[1] There are four common LiH isotopes with significantly different reduced mass, which allow a detailed investigation of isotopic effects in the breakdown of the Born–Oppenheimer approximation.[2]

The spectra of the $A\,^1\Sigma^+$–$X\,^1\Sigma^+$ system of the isotopic lithium hydrides were first observed by Nakamura in 1930[3] and 1931.[4] Emission and absorption spectra of ^{7}LiH were taken in the range from 3000 to 4100 Å, covering $v'' = 0$ to 2 and $v' = 1$ to 15. The system was further investigated by Crawford and Jorgensen[5] in 1935. They obtained some highly resolved absorption spectra of ^{7}LiH and ^{7}LiD in the region from 3200 to 4300 Å. In their study, a total of 26 bands covering v'' from 0 to 3 and v' from 0 to 14 for ^{7}LiH, and 35 bands covering v'' from 0 to 2 and v' from 1 to 18 for ^{7}LiD were reported. In terms of energy, Crawford and Jorgensen's data, however, only covered the lowest 23% of the $X\,^1\Sigma^+$ potential. Subsequent studies of the emission spectra of the LiH systems by Li[6], and Orth and Stwalley[7] in 1976 and 1979, respectively, further extended the data field of the $X\,^1\Sigma^+$ state to $v'' = 12$, and defined the lowest 70% of the LiH ground state potential.

In a previous study, Vidal and Stwalley[2] employed a variational method known as the inverted perturbation approach (IPA) to extract isotopic effects and accurate potential curves of the $X\,^1\Sigma^+$ ground state of the lithium hydrides. However, due to the lack of data for the higher vibrational levels, their calculations were limited to $v'' \leqslant 14$ for LiH, and $v'' \leqslant 8$ for LiD.

Recent studies of laser-induced fluorescence of the LiH isotopes, however, have revealed long progressions. For example, Ennen and co-workers[8] have recorded fluorescence transitions up to a vibrational level with $v'' = 26$, which corresponds to 96% of the ground state dissociation energy. Investigations at the University of Iowa by Verma and Stwalley[9] and also by Harding[10] discovered long laser-induced fluorescence (LIF) series of P, R doublets in ^{6}LiH, ^{7}LiH, and ^{7}LiD excited by ultraviolet argon ion laser lines at 3336, 3345, and 3358 Å, respectively. The LIF measurements of ^{6}LiH, in particular, cover over 99% of the $X\,^1\Sigma^+$ potential.

These LIF studies extended the observed data field very close to the dissociation limits and covered a region of the internuclear distance beyond 3.9 Å where the ionic-covalent curve crossing is believed to occur.[11,12] It is therefore possible to extend the IPA calculations and to examine in more detail phenomena such as the breakdown of the Born–Oppenheimer approximation and to study adiabatic and nonadiabatic effects. Unfortunately the LIF measurements have previously been recorded on strip charts. The accuracy of such a detection technique is on the order of $\pm$ 1 Å. Depending on the spectral region, this may correspond to errors of 2 to 10 cm^{-1}. The uncertainties of these measurements are therefore too large for studying the isotopic effects, which are typically on the order of 1 cm^{-1}. Therefore the laser-induced fluorescence spectra of the A–X system of the LiH isotopes are reinvestigated with higher resolution. The resulting data are used to reexamine the isotopic effects in these molecules, and to obtain accurate potential energy curves for the lithium hydride isotopes almost up to the dissociation limits.

II. EXPERIMENTAL

The ultraviolet lines of an argon ion (Coherent CR-18) and of a krypton ion (Coherent CR-3000) laser were separated from one another with a prism outside the laser cavities, and were then directed into a heat pipe oven which contained molted lithium (^{6}Li or ^{7}Li depending on the experiments of interest). The isotopically pure lithium metals, used in this study, were purchased from Oak Ridge National Laboratory, with an isotopic purity of 95.5% for lithium-6, and 99.8% for lithium-7.

The heat pipe was normally set at ~1000 K, which corresponds to a total lithium vapor pressure of about 1 Torr. A

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continuous flow of hydrogen (99.999% pure, purchased from Matheson Gas Products) or deuterium (99.97% pure, purchased from Air Products and Chemicals, Inc.) was maintained through the lithium vapor to produce the desired hydride/deuteride.

For obtaining the LIF spectra, photoelectric as well as photographic detection was used. Initially, the fluorescence lines were recorded photoelectrically through the perpendicular arm of the heat pipe, which had the shape of a cross. A half-meter Jarrell–Ash spectrometer was used which was equipped with a grating (1180 grooves/mm, blazed at 6000 Å) and a photomultiplier tube [S20 (EMI 9569 QB) or GaAs PMT (RCA 4832)]. Signal outputs of the PMT were amplified by a PAR 134 electrometer, and recorded on a strip chart recorder (Linseis LS44).

The three long LIF series of ^{6}LiH (3336 Å), ^{7}LiH (3345 Å), and ^{7}LiD (3358 Å) were recorded photographically with a 3.4 m Ebert spectrograph which was equipped with a grating of 7500 grooves/in. blazed at 37 000 Å. Kodak 103a-*O*, *G*, *D*, *F*, and *N* spectroscopic plates were used for detection. Unwanted signals from other orders were eliminated by suitable Corning color filters. Spectral lines of argon, krypton, mercury, neon, thorium, and xenon emitted from spectroscopic pen lamps and hollow cathode lamps, were used as spectral standards for the calibration of the fluorescence line positions. Spectrograms were taken in the fifth or the sixth order, with a dispersion of ~1.8 and ~1.2 Å/mm, respectively. For each of the three LIF spectra, six to seven exposures were necessary to cover the spectral regions of interest. Depending on the intensity of the fluorescence lines and the sensitivity of the spectroscopic plates, the exposure time for each spectrogram ranged from 2 to 48 h. As a result, 53 lines of the three different LIF series were recorded in the range from 4400 to 8300 Å. The line positions on the plates were measured at the Argonne National Laboratory using a manually operated comparator equipped with a photoelectric setting device. Six of the weaker lines, together with the final six lines of the ^{6}LiH (3336 Å) spectrum previously measured by Verma and Stwalley,[9] were measured with a manual comparator at the University of Iowa. Each plate was measured three to six times. A least squares program fitting a second order polynomial was used to interpolate the fluorescence line positions from the calibration standards. The line positions from different measurement varied by less than 0.02 cm^{-1}. In overlapping regions the absolute position of the fluorescence lines agreed to within 0.03 cm^{-1}.

III. RESULTS

The measurements performed in this study are summarized in Table I. Three new LIF transitions were identified, and three additional series were recorded photographically. For identifying spectral series, molecular constants from Vidal and Stwalley[2] were used.

Transition wavelengths (λ_{air} in Å) and vacuum frequencies (cm^{-1}) of three new LIF series of ^{6}LiD and ^{7}LiD, excited by the 3507 and 3564 Å lines of the Kr ion laser, have been obtained.[10] Due to their relatively weak fluorescence signals and short progressions, we did not make the attempt to record these spectra with the more accurate 3.4 m spectrograph. We also obtained fluorescence series of ^{6}LiH, ^{7}LiH, and ^{7}LiD, excited by the 3336, 3345, and 3358 Å argon ion laser lines, respectively.[10] Fluorescence lines below 4400 Å for the three series were not taken, since they have already been reported in the previous studies. In Fig. 1, the observed data fields of the three photographic measurements are compared with the previous spectroscopic data. The LIF spectra from these studies extended almost all the way up to the dissociation limits. In particular, the highest vibrational level observed for ^{7}LiD exceeds the previous spectroscopic plate measurements by 23 levels. In connection with the hydride data, where the highest levels observed in fluorescence are also considerably above the previously reported values, it has been possible to construct for the first time a complete set of ground state potential energy curves for both the hydride and deuteride extending close to the dissociation limits, and to examine the isotopic effects within the adiabatic or the Born–Oppenheimer approximations. These calculations are discussed in the following sections.

IV. MOLECULAR CONSTANTS

In this section, an extended set of molecular constants for the $X\ ^1\Sigma^+$ and the $A\ ^1\Sigma^+$ states are reported. The calculational procedures are similar to those used in Ref. 2.

In addition to the spectrographic results obtained in this work, the measurements by Crawford and Jorgensen,[5] Li

TABLE I. Laser coincidences of LiH isotopes investigated in this work.

Species	λ_{laser} (Å)	Observed range (Å)	Exciting transition (v', J' v'', J'')	v''_{max}	Detection method
^{6}LiH	3336	4500–8200	(12,10←0, 9)[c]	21	I[a]
^{7}LiH	3345	4500–8050	(16, 9←1, 10)[d]	22	I
^{7}LiD	3358	4400–8200	(19,22←0, 23)[d]	27	I
^{6}LiD	3507	3300–8200	(15,17←1, 16)	23	II[b]
^{7}LiD	3507	3300–6300	(13, 4←1, 3)	16	II
^{7}LiD	3564	3300–6400	(15, 6←2, 5)	17	II

[a] I Photographic (3.4 m).
[b] II Photoelectric (0.5 m).
[c] Assigned by Verma and Stwalley (Ref. 9).
[d] Assigned by Harding (Ref. 10).

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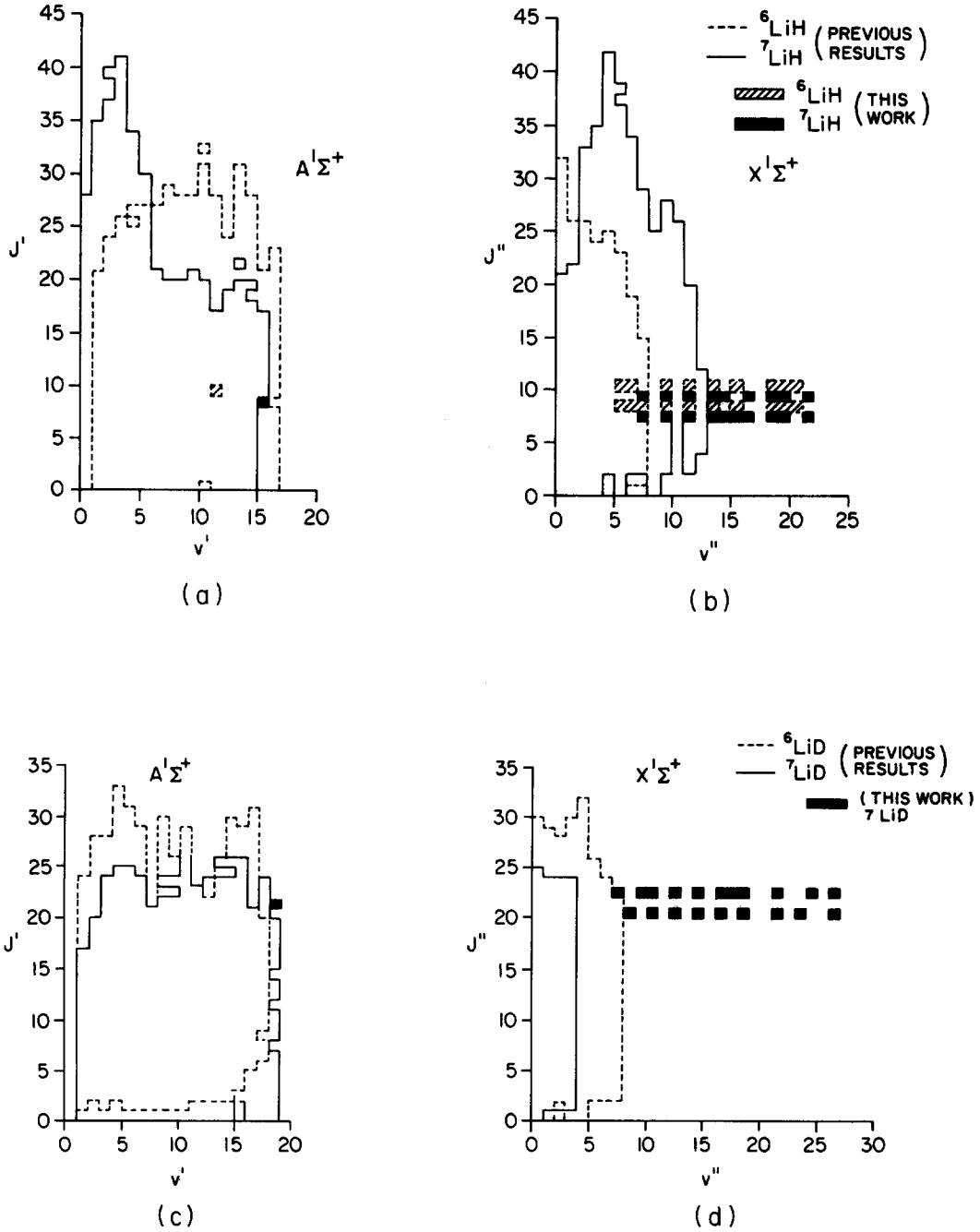


FIG. 1. Comparisons of the ranges of v and J levels observed in the present LIF spectra with previous spectrographic measurements: (a) $A\ ^1\Sigma^+$ of LiH; (b) $X\ ^1\Sigma^+$ of LiH, (c) $A\ ^1\Sigma^+$ of LiD; and (d) $X\ ^1\Sigma^+$ of LiD.

and Stwalley,[6] and Orth and Stwalley[7] have been included in fitting the molecular constants. Microwave measurements of the two lowest vibrational levels of the isotopic lithium hydrides by Pearson and Gordy,[13] and Plummer *et al.*[14] were also incorporated, to improve the rotational constants of the ground state electronic state. The data fields of vibrational and rotational values used in this calculation are summarized in Fig. 1.

The observed line positions for transitions between the vibrational–rotational levels of the excited state (v',J') and of the ground state (v'',J''), are given by

$$\nu(v',J';v'',J'') = T'(v',J') - T''(v'',J''). \quad (1)$$

The term values T are most conveniently expressed by means of a Dunham-type expansion[2]:

$$T(v,J) = \sum_{i,k} \overline{A}_{ik}(v+1/2)^i[J(J+1)]^k \mu_r^{-i/2-k}, \quad (2)$$

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where

$$\bar{A}_{ik} = A_{ik} + B^{\mathrm{H}}_{ik}/M_H + B^{\mathrm{Li}}_{ik}/M_{\mathrm{Li}} \quad (3)$$

and

$$\mu_r = \mu/\mu_i. \quad (4)$$

In Eq. (3), A_{ik} is a mass invariant parameter, while B^{H}_{ik} and B^{Li}_{ik} describe the corrections for the mass contributions of H and Li, respectively. μ_r is the relative reduced mass, with i denoting the reference isotopic species ^{7}LiH.

The masses of the H, D, ^{6}Li, and ^{7}Li atoms in units of $M(^{12}C)/12$ have been taken from Wapstra and Bos.[15] They were given in Table I of Ref. 2.

TABLE II. The Dunham-type coefficients of the $X\,^1\Sigma^+$ state and their adiabatic corrections as obtained from a least squares fit of the measured line positions. (All values are given in cm^{-1}, together with their standard errors).

A''_{ik}	0	1	2	3	4	5
0		7.520 949 (0.000 076)	− 8.614 061 − 4 (0.002 5 − 4)	1.074 82 − 7 (0.004 9 − 7)	1.695 92 − 11 (0.052 5 − 11)	1.858 04 − 15 (0.156 5 − 15)
1	1405.680 97 (0.149 4)	− 2.195 880 − 1 (0.000 33)	1.609 64 − 5 (0.018 1 − 5)	− 2.841 86 − 10 (1.318 2 − 10)		
2	− 24.163 81 (0.208 3)	6.355 71 − 3 (0.507 − 3)	− 1.858 19 − 7 (0.514 8 − 7)	− 1.534 48 − 10 (0.129 6 − 10)		
3	0.825 86 (0.137 9)	− 2.839 03 − 3 (0.372 7 − 3)	2.282 99 − 8 (0.873 40 − 8)			
4	− 0.225 46 (0.048 7)	9.465 53 − 4 (1.401 4 − 4)	− 2.912 05 − 9 (0.414 7 − 9)			
5	4.198 66 − 2 (9.634 − 3)	− 1.770 38 − 4 (0.288 30 − 4)				
6	− 4.477 8 − 3 (1.088 − 3)	1.875 52 − 5 (0.333 97 − 5)				
7	2.675 5 − 4 (0.692 0 − 4)	− 1.119 0 − 6 (0.216 1 − 6)				
8	− 8.370 3 − 6 (2.295 − 6)	3.499 3 − 8 (0.725 6 − 8)				
9	(1.064 2 − 7 (3.077 − 8)	− 4.475 5 − 10 (0.981 3 − 10)				
B^{H}_{ik}	0	1	2			
0		− 7.535 507 − 3 (0.12 − 3)	3.420 69 − 6 (0.43 − 6)			
1	− 1.615 702 (0.18)	4.358 47 − 3 (0.45 − 3)	− 4.079 96 − 7 (2.0 − 7)			
2	1.495 073 (0.23)	− 5.523 48 − 3 (0.58 − 3)				
3	− 8.927 767 − 1 (1.4 − 1)	3.349 36 − 3 (0.39 − 3)				
4	2.816 06 − 1 (0.50 − 1)	− 1.073 86 − 3 (0.14 − 3)				
5	− 5.010 69 − 2 (0.97 − 2)	1.936 21 − 4 (0.29 − 4)				
6	5.133 2 − 3 (1.1 − 3)	− 2.001 6 − 5 (0.34 − 5)				
7	− 2.988 0 − 4 (0.07 − 4)	1.171 7 − 6 (0.22 − 6)				
8	9.167 0 − 6 (2.3 − 6)	− 3.607 5 − 8 (0.73 − 8)				
9	− 1.151 5 − 7 (0.31 − 7)	4.543 2 − 10 (0.99 − 10)				
B^{Li}_{ik}	0	1				
0		− 5.415 91 − 4 (0.10 − 4)				

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TABLE III. The Dunham-type coefficients of the $A\ ^1\Sigma^+$ state and their adiabatic corrections as obtained from a least squares fit of the measured line positions. (All values are given in cm^{-1}, together with their standard errors).

A'_{ik}	0	1	2	3	4	5	6
0	26 514.020 (0.17)	2.842 782 (0.0015)	− 1.406 085 71 − 3 (0.0038 − 3)	1.429 757 63 − 6 (0.009 0 − 6)	− 7.850 885 − 10 (0.10 − 10)	7.015 216 − 14 (0.61 − 14)	5.568 800 − 17 (0.21 − 17)
1	241.491 91 (0.30)	6.850 432 − 2 (0.25 − 2)	6.363 480 9 − 4 (0.069 − 4)	− 1.489 021 56 − 6 (0.107 − 6)	1.207 110 − 9 (0.017 − 9)	− 2.979 482 − 13 (0.059 − 13)	− 1.742 316 − 17 (0.096 − 17)
2	25.587 372 (0.21)	− 3.480 526 − 2 (0.16 − 2)	− 1.876 996 24 − 4 (0.045 − 4)	6.256 472 − 7 (0.11 − 7)	− 5.771 949 − 10 (0.11 − 10)	1.611 102 − 13 (0.037 − 13)	9.105 405 − 19 (1.1 − 19)
3	− 3.386 070 7 (0.077)	1.178 374 − 2 (0.054 − 2)	2.820 269 0 − 5 (0.13 − 5)	− 1.352 106 − 7 (0.030 − 7)	1.307 365 − 10 (0.029 − 10)	− 3.532 629 − 14 (0.096 − 14)	
4	0.438 714 56 (0.017)	− 2.568 172 − 3 (0.11 − 3)	− 1.803 740 − 6 (0.21 − 6)	1.635 034 − 8 (0.046 − 8)	− 1.590 965 − 11 (0.041 − 11)	3.826 562 − 15 (0.12 − 15)	
5	− 4.644 153 − 2 (0.24 − 2)	3.379 276 − 4 (0.14 − 4)	− 4.098 751 8 − 8 (1.9 − 8)	− 1.112 403 − 9 (0.037 − 9)	1.058 473 − 12 (0.031 − 12)	− 2.024 114 − 16 (0.075 − 16)	
6	3.348 933 − 3 (0.21 − 3)	− 2.681 964 − 5 (0.11 − 5)	1.293 357 3 − 8 (0.10 − 8)	3.963 238 − 11 (0.15 − 11)	− 3.588 959 − 14 (0.12 − 14)	4.164 846 − 18 (0.18 − 18)	
7	− 1.495 705 − 4 (0.12 − 4)	1.237 428 − 6 (0.059 − 6)	− 6.951 754 9 − 10 (0.30 − 10)	− 5.726 916 − 13 (0.26 − 13)	4.777 065 − 16 (0.18 − 16)		
8	3.646 056 − 6 (0.37 − 6)	− 2.978 858 − 8 (0.18 − 8)	1.250 897 8 − 11 (0.045 − 11)				
9	− 3.600 946 − 8 (0.50 − 8)	2.748 831 − 10 (0.23 − 10)					
B^H_{ik}	0	1	2				
0	2.391 289 (0.16)	− 4.228 012 − 2 (0.13 − 2)	− 5.284 6580 − 6 (1.2 − 6)				
1	− 6.964 161 (0.26)	5.270 478 − 2 (0.20 − 2)	2.461 445 − 6 (0.35 − 6)				
2	3.251 369 (0.16)	− 2.746 700 − 2 (0.12 − 2)	− 1.696 747 − 7 (0.23 − 7)				
3	− 0.823 5608 (0.049)	7.227 034 − 3 (0.35 − 3)					
4	0.117 7298 (0.008 2)	− 1.056 230 − 3 (0.056 − 3)					
5	− 9.487 394 − 3 (0.76 − 3)	8.667 768 − 5 (0.51 − 5)					
6	4.009 441 − 4 (0.36 − 4)	− 3.728 937 − 6 (0.24 − 6)					
7	− 6.900 888 − 6 (0.69 − 6)	6.536 019 − 8 (0.45 − 8)					
B^H_{ik}	0	1					
0	1.917 (0.25)	− 5.430 − 3 (0.76 − 3)					
1	− 5.162 − 1 (0.29 − 1)						

In performing the global least squares fit of the molecular constants A_{ik} and B_{ik} in Eq. (2), a weighted least squares program was used.[2] The weights of the different sets of data were taken from the root mean square (rms) difference of the observed and the calculated values, and were established after a few iterations. Statistically insignificant data points were eliminated if they had only a 99% level of confidence. In this manner, a total number of 664 lines, including one of the LIF points, were eliminated from the data sets. These data corresponded to 8% of the total number of measured line positions. Examining the eliminated data, one finds that a large fraction of them were classified as blended in the original measurements. Most of the other eliminated data more or less scattered over the entire data field. It is therefore clear that the large deviations of these data are most likely due to inaccurate measurements, or due to erroneous assignments. Finally, 7487 lines were incorporated in the global least squares fit.

The results of the final fit are given in Table II for the $X\ ^1\Sigma^+$ state and in Table III for the $A\ ^1\Sigma^+$ state. The molecular constants and their standard errors are given in units of cm^{-1}. A total number of 52 A_{ik} and B_{ik} constants are used to define the ground state term values, as compared to 23 used in Ref. 2. It is, however, not surprising that more terms have to be included in this study to account for the anharmonicity of the levels which are close to the dissociation limit. Similar

TABLE IV. Root mean square (rms) errors of the different data sets used in the least squares fit program.

Species	N_{total}	rms
^{6}LiH	2242	0.128
^{7}LiH	2313	0.056
^{6}LiD	1711	0.125
^{7}LiD	1153	0.028
Microwave	10	8.3E − 7
This study	58	0.202

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TABLE V. List of J values used in the variational method for generating the IPA potentials. (The J values are followed in brackets by the range of vibrational quantum numbers employed.)

^{6}LiH:	2,6(0–7), 9,11(0–8,10,12,14,16,19,20,21), 14,17(0–7), 20(0–6), 23,26(0–5), 28(0–1), 30(0)
^{7}LiH:	2,6(0–12), 8(0–12,14–17,19,22), 10(0–12,14,15,17,19,20,22), 14,17,20(0–11), 23(0–10), 25(0–9), 32(0–6), 40(2–3)
^{6}LiD:	2,6,10,14,17,23(0,5), 26(0–5), 29(0–4)
^{7}LiD:	2,6,10,14,17(0–3), 21(0–4,9,11,13,15,17,19,22,24,27), 23(0–4,8,10,11,13,15,17–19,22,25,27), 24(0–3)

to the results reported in Ref. 2, the mass dependence of the molecular constants corresponding to the lithium isotopes is almost negligible. On the other hand, a large number of terms had to be added to describe the effects of interchanging a hydrogen and a deuterium atom.

Table IV shows the standard errors for the different subsets of data used in the least squares fit. In particular, the rms error of the microwave data is 8.3×10^{-7} cm^{-1}, which accurately defines the rotational constants of the lowest vibrational levels. The total rms error of this study is 0.202 cm^{-1}, which is somewhat large; however, in view of the large anharmonicity of the high vibrational levels which have been incorporated in the data set, the rms value is not unreasonable.

V. CALCULATION OF IPA POTENTIALS

In this section the IPA potentials of the electronic ground state and their adiabatic corrections are examined for the four different isotopic combinations of LiH.

Within the adiabatic approximation, the potential energy consists of two terms:

$$U(R) = U_{\mathrm{BO}}(R) + \Delta U_{\mathrm{Ad}}(R), \tag{5}$$

where $U_{\mathrm{BO}}(R)$ is the nuclear mass independent component, better known as the Born–Oppenheimer potential. The other term, denoted by $\Delta U_{\mathrm{Ad}}(R)$, is the adiabatic correction to the mass invariant potential $U_{\mathrm{BO}}(R)$. The $U_{\mathrm{Ad}}(R)$ can be further broken up into two components[2]:

$$U_{\mathrm{Ad}}(R) = [\Delta U^{\mathrm{A}}_{\mathrm{Ad}}(R)]/M_{\mathrm{A}} + [\Delta U^{\mathrm{B}}_{\mathrm{Ad}}(R)]/M_{\mathrm{B}}. \tag{6}$$

With these relations, one can separately examine the adiabatic contributions due to the two component elements A and B.

The methods of calculation have already been explained in detail in Sec. III of Ref. 2 and are not repeated here. The interested reader is therefore referred to that paper.

In the IPA calculations one should, in principle, include all the measured term values to simultaneously generate both the upper and lower state IPA potentials. However, this may not be very practical. In the case of LiH, one would have to solve nearly 15 000 second order differential equations for just one iteration in order to incorporate all 7487 measured fluorescent lines. In addition, much of the information is redundant and this technique would be very inefficient. Therefore, the term values derived for the least squares fit of the experimental data were used in the calculations. A number of representative levels distributed over different areas of the data fields were used to represent the data sets. In this manner, the computational effort was greatly reduced. The term values selected in the variational method are summarized in Table V, which gives the rotational levels J for a particular isotopic combination followed in brackets by the range of vibrational quantum numbers.

It should be mentioned that the highest vibrational levels of the P series observed in ^{7}LiH and ^{7}LiD are quasibound, giving rise to calculated half-widths of 0.2 and 2.2×10^{-8} cm^{-1} for the (v,J) levels (22,10) ^{7}LiH, and (27,23) ^{7}LiD, respectively. In order to include these two highest levels, an arbitrary potential barrier was imposed at large internuclear separations (i.e., $R > 7.3$ Å). In this manner, the quasibound levels became bound, and the corresponding Schrödinger equation could be solved with the usual numerical integration method for a bound state. The accuracy of the calculations is not affected by the artificial potential barrier at 7.3 Å, since the barrier is sufficiently far away.

TABLE VI. The IPA potential energy values of the ^{6}LiH $X\ ^1\Sigma^+$ state.[a]

v	G_v(cm^{-1})	R_-(Å)	R_+(Å)
0	705.08	1.444 898	1.779 146
1	2 078.45	1.352 045	1.938 830
2	3 406.08	1.295 452	2.063 742
3	4 688.81	1.253 444	2.175 611
4	5 927.55	1.219 820	2.280 749
5	7 123.12	1.191 788	2.382 091
6	8 276.16	1.167 807	2.481 340
7	9 387.21	1.146 922	2.579 639
8	10 456.61	1.128 490	2.677 848
9	11 484.55	1.112 049	2.776 697
10	12 470.99	1.097 255	2.876 878
11	13 415.62	1.083 840	2.979 132
12	14 317.74	1.071 587	3.084 332
13	15 176.19	1.060 318	3.193 580
14	15 989.12	1.049 885	3.308 349
15	16 753.86	1.040 180	3.430 690
16	17 466.64	1.031 188	3.563 582
17	18 122.33	1.023 169	3.711 551
18	18 713.93	1.017 210	3.881 905
19	19 232.10	1.012 766	4.087 471
20	19 664.54	1.009 039	4.352 574
21	19 995.83	1.006 196	4.732 532

[a] $R_e = 1.595\ 584$ Å.

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TABLE VII. The IPA potential energy values of the ^{7}LiH $X\,^1\Sigma^+$ state.[a]

v	G_v(cm^{-1})	R_-(Å)	R_+(Å)
0	697.88	1.445 602	1.778 101
1	2 057.59	1.353 100	1.936 731
2	3 372.48	1.296 686	2.060 731
3	4 643.37	1.254 789	2.171 721
4	5 871.14	1.221 239	2.275 980
5	7 056.58	1.193 255	2.376 423
6	8 200.35	1.169 306	2.474 742
7	9 302.95	1.148 440	2.572 066
8	10 364.73	1.130 017	2.669 243
9	11 385.90	1.113 579	2.766 987
10	12 366.42	1.098 783	2.865 968
11	13 306.04	1.085 363	2.966 897
12	14 204.13	1.073 105	3.070 600
13	15 059.61	1.061 831	3.178 112
14	15 870.80	1.051 394	3.290 801
15	16 635.24	1.041 682	3.410 559
16	17 349.46	1.032 660	3.540 102
17	18 008.73	1.024 504	3.683 509
18	18 606.62	1.018 195	3.847 256
19	19 134.53	1.013 582	4.042 445
20	19 581.14	1.009 757	4.289 738
21	19 932.13	1.006 741	4.633 121
22	20 169.84	1.004 707	5.204 851

[a] $R_e = 1.595\,584$ Å.

Finally, 14, 3, and 1 different $f_i(R)$ functions were used, in the variational procedure, to obtain the potential energies $U_{BO}(R)$, $\Delta U^{H}_{Ad}(R)$, and $\Delta U^{Li}_{Ad}(R)$, respectively. Starting with the RKR potential of ^{7}LiH, the rms error of the energy difference E_{vJ} for the $X\,^1\Sigma^+$ state of the four isotopic LiH molecules was initially as large as 3.37 cm^{-1}. This error, however, rapidly went down to 0.08 cm^{-1} after only three iterations.

In Tables VI–IX, the IPA potentials of the isotopic LiH molecules are given, which correspond to the energy levels of the vibrational levels with $J = 0$. It however has to be pointed out that the inner turning points R_- of the highest v'' levels of the three IPA potentials are outside the experimentally defined regions, and therefore are not as significant as the rest of the data.

At large internuclear separations, $R = 7$ Å, the potentials of the ground state IPA data extrapolated to the values

TABLE VIII. The IPA potential energy values of the ^{6}LiD $X\,^1\Sigma^+$ state.[a]

v	G_v(cm^{-1})	R_-(Å)	R_+(Å)
0	534.29	1.462 644	1.752 775
1	1581.75	1.379 025	1.886 663
2	2602.88	1.327 237	1.989 568
3	3598.00	1.288 318	2.080 461
4	4567.51	1.256 819	2.164 822
5	5511.80	1.230 275	2.245 166
6	6431.25	1.207 330	2.322 909
7	7326.17	1.187 142	2.398 959
8	8196.84	1.169 150	2.473 951
9	9043.48	1.152 954	2.548 366

[a] $R_e = 1.595\,304$ Å.

TABLE IX. The IPA potential energy values of the ^{7}LiD $X\,^1\Sigma^+$ state.[a]

v	G_v(cm^{-1})	R_-(Å)	R_+(Å)
0	524.69	1.463 749	1.751 219
1	1 553.71	1.380 713	1.883 601
2	2 557.33	1.329 236	1.985 242
3	3 535.86	1.290 523	2.074 944
4	4 489.67	1.259 170	2.158 140
5	5 419.13	1.232 733	2.237 320
6	6 324.60	1.209 867	2.313 886
7	7 206.40	1.189 737	2.388 733
8	8 064.77	1.171 786	2.462 489
9	8 899.95	1.155 619	2.535 624
10	9 712.08	1.140 942	2.608 516
11	10 501.29	1.127 529	2.681 485
12	11 267.63	1.115 204	2.754 823
13	12 011.09	1.103 823	2.828 809
14	12 731.57	1.093 267	2.903 736
15	13 428.90	1.083 436	2.979 918
16	14 102.75	1.074 243	3.057 722
17	14 752.68	1.065 614	3.137 578
18	15 378.05	1.057 484	3.220 023
19	15 978.00	1.049 796	3.305 726
20	16 551.42	1.042 509	3.395 552
21	17 096.85	1.035 604	3.490 637
22	17 612.46	1.029 121	3.592 516
23	18 095.94	1.023 244	3.703 313
24	18 544.37	1.018 590	3.826 083
25	18 954.15	1.014 931	3.965 413
26	19 320.78	1.011 822	4.128 586
27	19 638.84	1.009 080	4.327 463

[a] $R_e = 1.595\,304$ Å.

of 20 285.1 and 20 290.8 cm^{-1} for LiH and LiD, respectively. As compared with the recommended dissociation energies by Vidal and Stwalley[16] (based on data for the excited $B\,^1\Pi$ state), the differences are less than 3 cm^{-1}. However, the extrapolations of Ref. 16 are more accurate and should still be considered as the better values.

The IPA calculations show that the contribution of the isotopic Li atom to the adiabatic correction function for the ground state potential energy is almost insignificant. On the other hand, interchanging a hydrogen isotope in the LiH molecule causes a systematic shift of the potential.

The adiabatic correction function $\Delta U^{H}_{Ad}(R)$ is tabulated in Table X. It is compared with the results of the IPA analysis of Vidal and Stwalley[2] and with the *ab initio* calculations of Bishop and Cheung[1] in Fig. 2. In comparison with the adiabatic correction calculated by Vidal and Stwalley, the two IPA calculations agree with each other very well over the region around the bottom of the potential curve. This has to be expected since, except for the LIF data used in this calculation, the same data sets were used in both studies. However, at larger R values, the two calculations deviate significantly from each other. The ΔU^{H}_{Ad} function calculated here levels off to about -12 cm^{-1}, which is quite reasonable, since at large R, the function is simply given by

$$\lim_{R\to\infty} \Delta U^{H}_{Ad}(R) \approx [D_e(\mathrm{LiH}) - D_e(\mathrm{LiD})]/M_H^{-1} - M_D^{-1}) \tag{7}$$

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TABLE X. The Born–Oppenheimer potential U_{BO} and the adiabatic correction ΔU^H_{Ad} and the standard error of the adiabatic correction, each given in cm^{-1}.

R(Å)	U_{BO}	ΔU^H_{Ad}	σ^H
1.058	15 314.07	35.32	0.63
1.069	14 495.56	33.92	0.59
1.081	13 619.53	32.44	0.55
1.094	12 691.35	30.87	0.54
1.108	11 715.70	29.20	0.51
1.124	10 696.20	27.44	0.47
1.142	9 635.18	25.57	0.44
1.162	8 533.78	23.59	0.40
1.185	7 392.06	21.47	0.37
1.212	6 209.36	19.18	0.33
1.244	4 984.53	16.69	0.28
1.284	3 716.29	13.93	0.24
1.336	2 403.54	10.73	0.18
1.415	1 045.83	6.66	0.12
1.596	− 0.02	0.04	0.01
1.722	357.98	− 3.22	0.03
1.904	1 746.27	− 6.47	0.06
2.034	3 086.94	− 8.11	0.07
2.148	4 382.00	− 9.23	0.08
2.255	5 632.59	− 10.05	0.08
2.357	6 839.60	− 10.69	0.09
2.457	8 003.76	− 11.20	0.09
2.555	9 125.67	− 11.61	0.09
2.653	10 205.74	− 11.94	0.09
2.752	11 244.28	− 12.21	0.09
2.852	12 241.38	− 12.43	0.09
2.953	13 196.87	− 12.61	0.09
3.058	14 110.19	− 12.75	0.10
3.166	14 980.29	− 12.87	0.10
3.279	15 805.46	− 12.96	0.10
3.400	16 583.13	− 13.02	0.11
3.530	17 309.68	− 13.06	0.11
3.674	17 980.16	− 13.08	0.12
3.837	18 587.85	− 13.07	0.12
4.032	19 123.75	− 13.03	0.13
4.418	19 754.07	− 12.89	0.15
4.859	20 072.73	− 12.67	0.17
5.080	20 151.23	− 12.55	0.18

which is equal to $-11.6\ cm^{-1}$. The curve in Fig. 2 shows a smooth extrapolation to that value at large R. In Ref. 2, the ΔU^H_{Ad} curve bends up beyond about 2.4 Å, which is however very close to the limit defined by the data incorporated in that calculation. This also demonstrates the importance of including the high lying levels in the IPA calculations. The *ab initio* calculations of ΔU^H_{Ad} by Bishop and Cheung, on the other hand, show a significant disagreement with the present result. In comparison with the IPA calculations, their calculated ΔU^H_{Ad} function is somewhat shallower. For instance, at $R = 2.4$ Å, the difference between the two results is about $4\ cm^{-1}$. The difference, however, increases with increasing R, which is due to the increase of their calculated values beyond $R = 2.4$ Å, as opposed to the leveling off observed in this study. The source of these discrepancies has not yet been settled. Bishop and Cheung[1] have suggested that our IPA analysis of the adiabatic corrections may contain, to some extent, nonadiabatic contributions, whereas the theoretical study gives only the genuine adiabatic corrections. If this were true, the difference between our results and their calculations should primarily give the nonadiabatic contributions. However, it is then difficult to understand why, for the electronic states of interest, the nonadiabatic terms should be comparable to the adiabatic corrections.

The appearance of the adiabatic correction function in Fig. 2 shows that the ΔU^H_{Ad} function is positive for $R < R_e$, and is negative for $R > R_e$, which indicates a systematic shift of the deuteride potential towards smaller internuclear distances. The positive and negative values of the ΔU^H_{Ad} are about the same order of magnitude, which results in a small isotopic difference of the energy values.

One other interesting feature of the adiabatic function around 3.9 Å should be noted where the crossing of the ionic-covalent zero order potential energy curves occurs. The function does not reveal any peculiar shape as noted in the $A\,^1\Sigma^+$ adiabatic correction curve.[2]

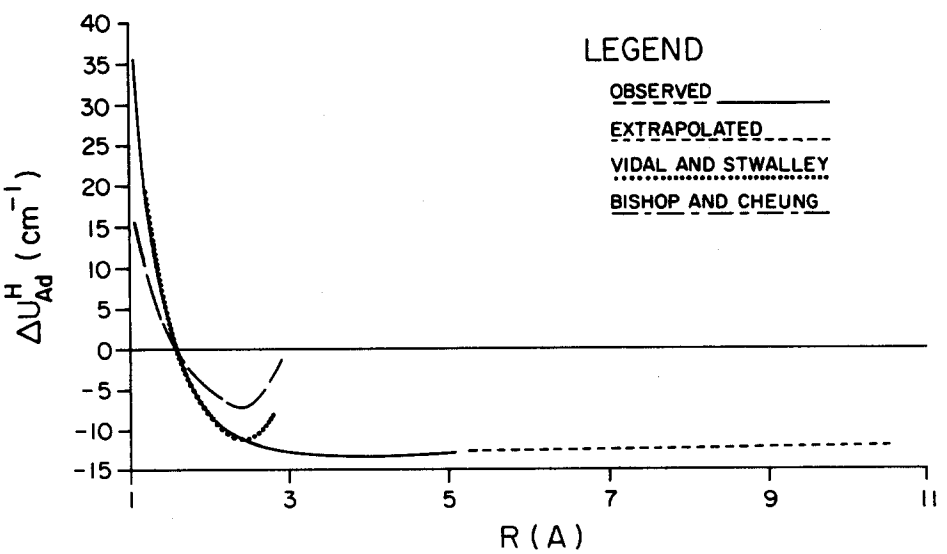


FIG. 2. The adiabatic correction of the $X\,^1\Sigma^+$ state due to the hydrogen isotopes as a function of the internuclear separation R.

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